HIGH BEAM CONTROL (HBC) SYSTEM

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Outline

- The High Beam Control (HBC) system performs automatic switching to low beams when the forward sensing camera (FSC) installed to the windshield recognizes a vehicle ahead and when traveling through towns and cities while the vehicle is being driven with the headlights in the high beam position. Due to this, blinding of other vehicles from headlight glare is prevented and driver visibility is assured.

Caution

- **Under the following conditions, the ability of the forward sensing camera (FSC) to detect light-emitting objects decreases, which could result in the system not operating normally and the inability to detect vehicles ahead.**
  - **— Brightness of moonlight in surrounding area**
  - **— Poor visibility due to rain and fog**
  - **— Scratches or dirt on the camera lens or windshield glass**
- **Do not rely solely on the High Beam Control (HBC) system, and perform the appropriate switching operation based on a visual determination of the conditions.**
- **If the forward sensing camera (FSC) aiming is not completed correctly, the forward sensing camera (FSC) cannot record the camera shooting angle and the High Beam Control (HBC) system may not operate normally. When performing the following servicing, always perform the forward sensing camera (FSC) aiming. In addition, for the forward sensing camera (FSC) aiming procedure, refer to the [FORWARD SENSING CAMERA (FSC) AIMING] in the Workshop Manual.**
  - **— Forward sensing camera (FSC) replacement**
  - **— Forward sensing camera (FSC) clip replacement**
  - **— Windshield replacement**

Function

- If the forward sensing camera (FSC) recognizes light-emitting objects (vehicles ahead, street lights (driving in town/city)) while the vehicle is being driven with the light switch in the AUTO and the dimmer switch in the HI position (headlights HI on condition), it automatically switches to low beams.
- For details of the forward sensing camera (FSC), refer to the [FORWARD SENSING CAMERA (FSC)]. (See FORWARD SENSING CAMERA (FSC).)

Setting function for control order of precedence

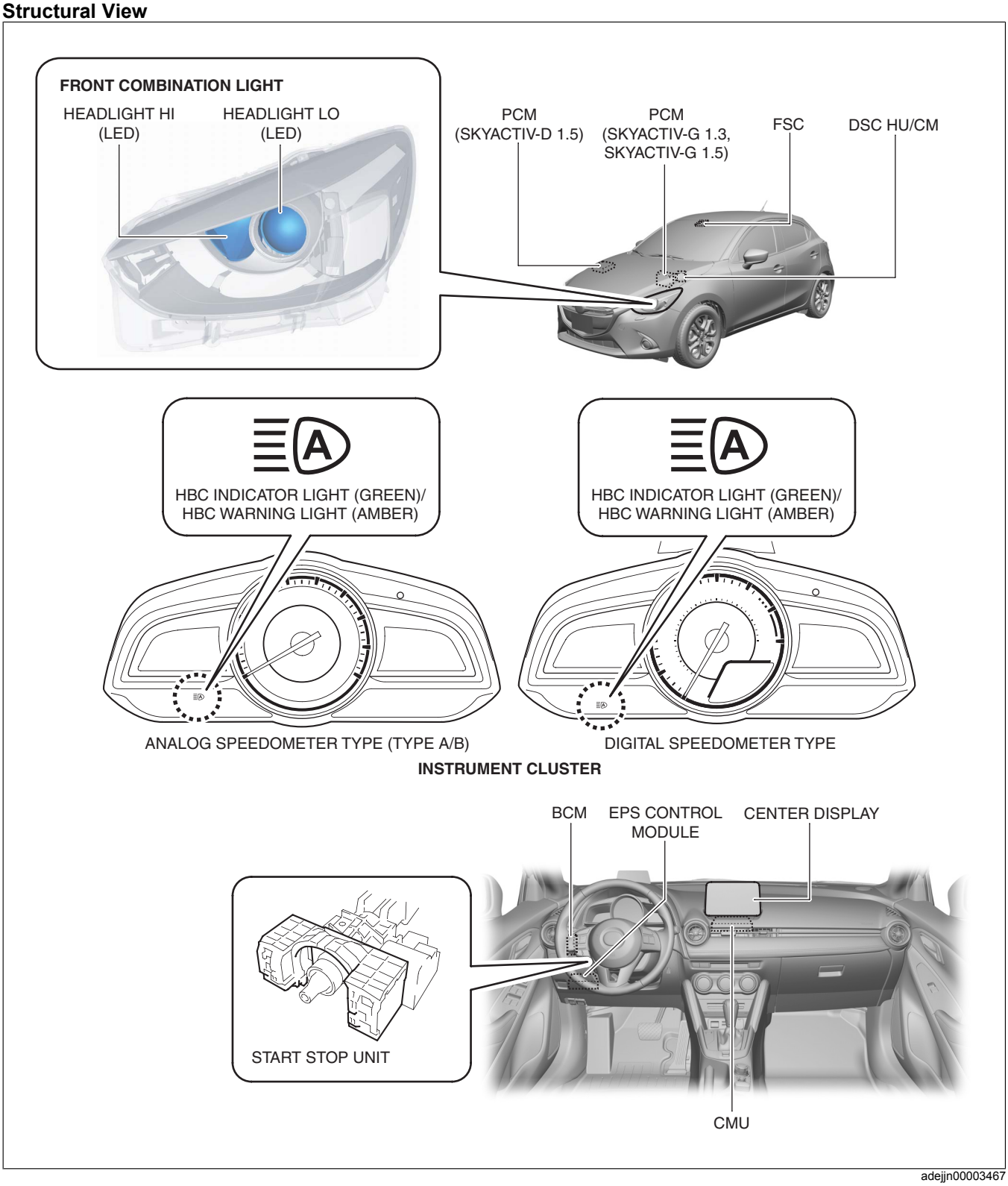
- If several light-emitting objects are recognized (vehicles ahead, street lights (driving in town/city)), control is conducted according to the following order of precedence.
  1. Closest on-coming vehicle
  2. Closest vehicle ahead (no on-coming vehicles)
  3. Street lights (driving in town/city)

System conditions display function

- The forward sensing camera (FSC) displays the system status using the HBC indicator light (green) and HBC warning light (amber) in the instrument cluster, and the warning display on the center display. For the warning display content on the center display, refer to the [CENTER DISPLAY]. (See CENTER DISPLAY [WITH CENTER DISPLAY].)

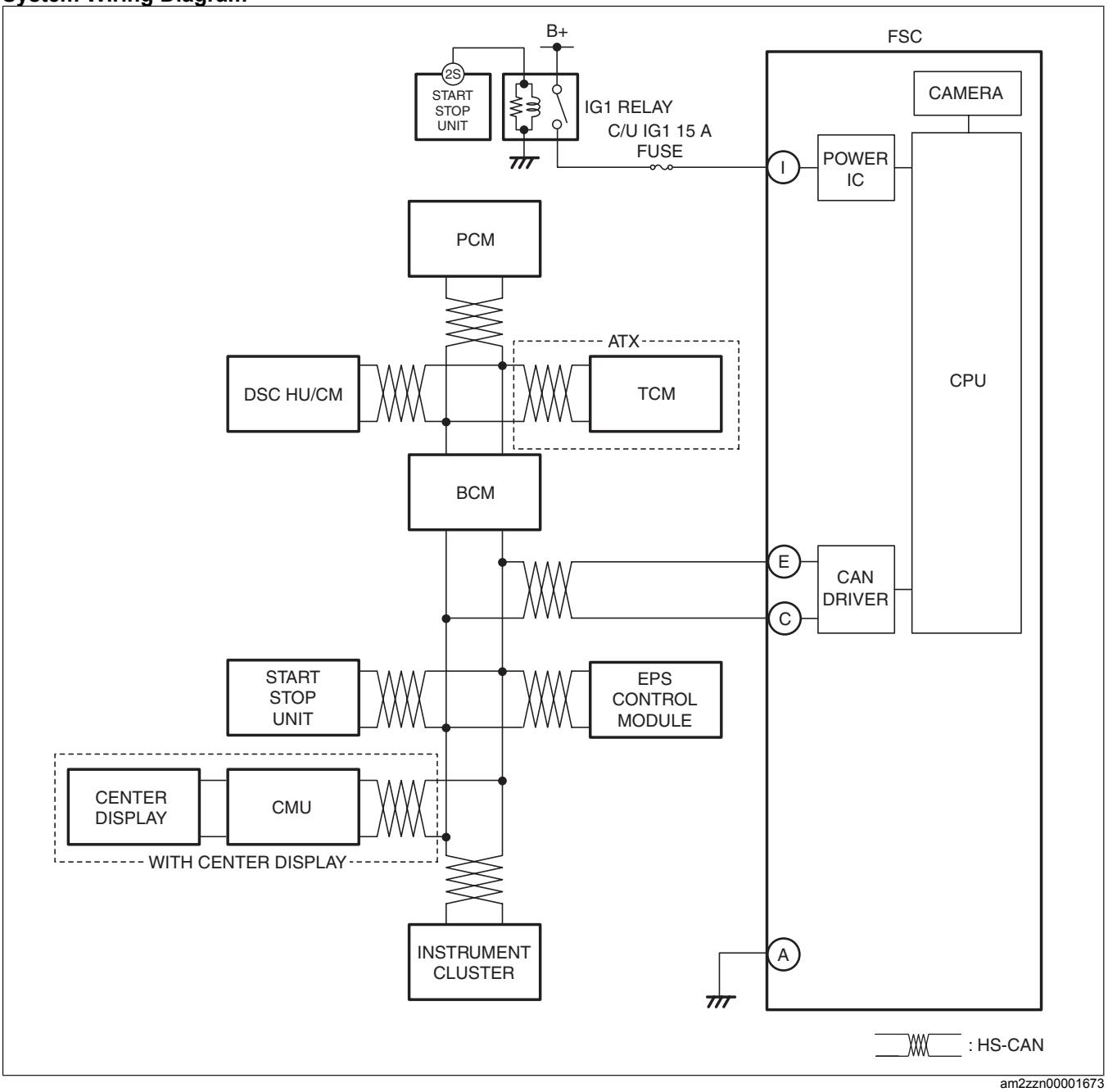
<table>
<tr><th>Condition</th><th>HBC indicator light (green)</th><th>HBC warning light (amber)</th><th>Center display</th></tr>
<tr><td>Light switch is in AUTO and dimmer switch is in position other than HI</td><td rowspan="4">Off</td><td rowspan="5">Off</td><td rowspan="5">No display</td></tr>
<tr><td>Headlights off</td></tr>
<tr><td>While vehicle is stopped</td></tr>
<tr><td>System set to off by personalization</td></tr>
<tr><td>Light switch is in AUTO and dimmer switch is in HI and headlights turned on</td><td>On</td></tr>
<tr><td>Forward sensing camera (FSC) detects camera fogging</td><td rowspan="4">Off</td><td rowspan="4">On</td><td rowspan="4">Warning display</td></tr>
<tr><td>Forward sensing camera (FSC) detects that the camera is dirty</td></tr>
<tr><td>Malfunction in HBC occurs</td></tr>
<tr><td>Malfunction in forward sensing camera (FSC) occurred</td></tr>
</table>

Structural View



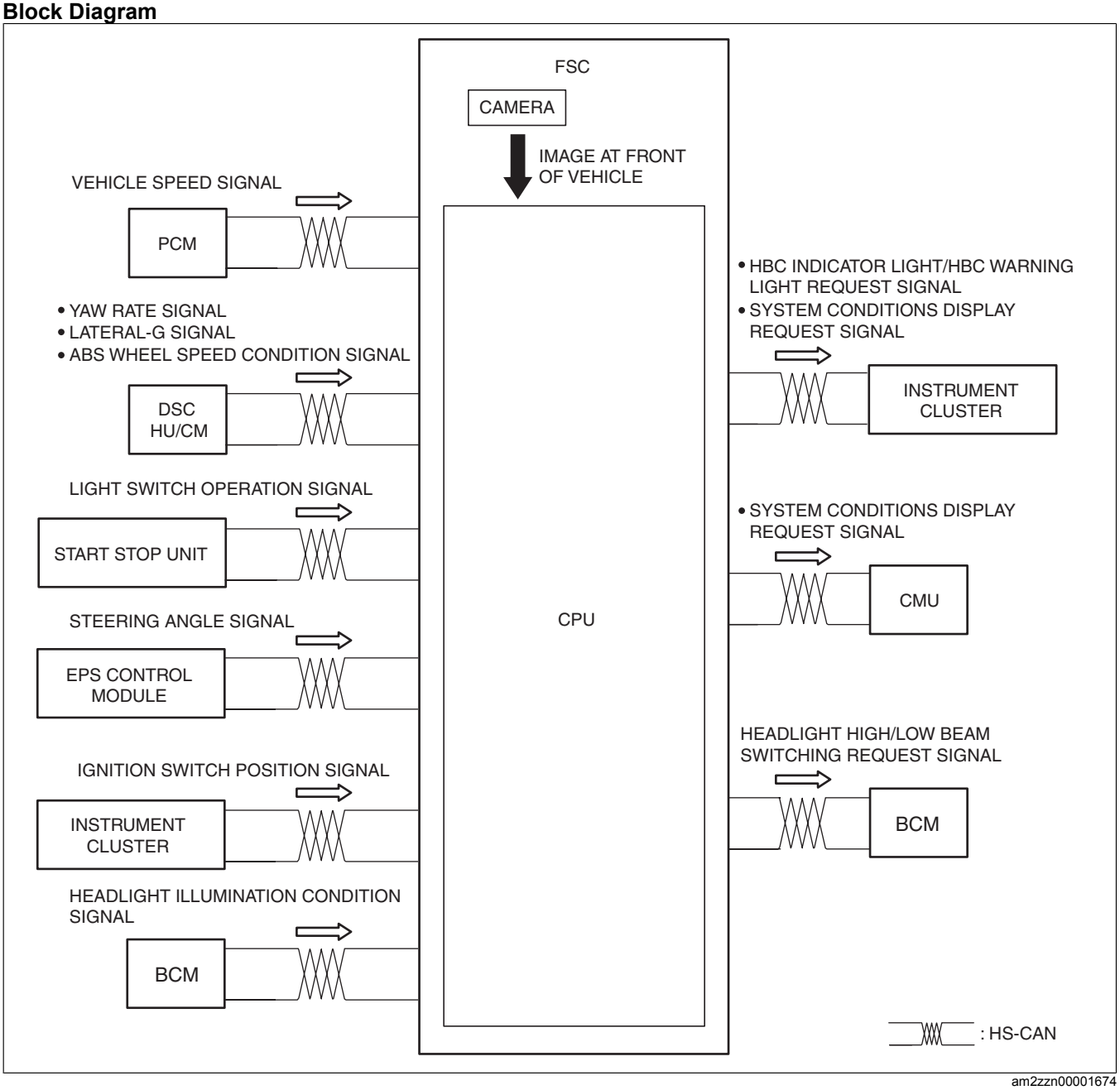
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System Wiring Diagram



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Block Diagram



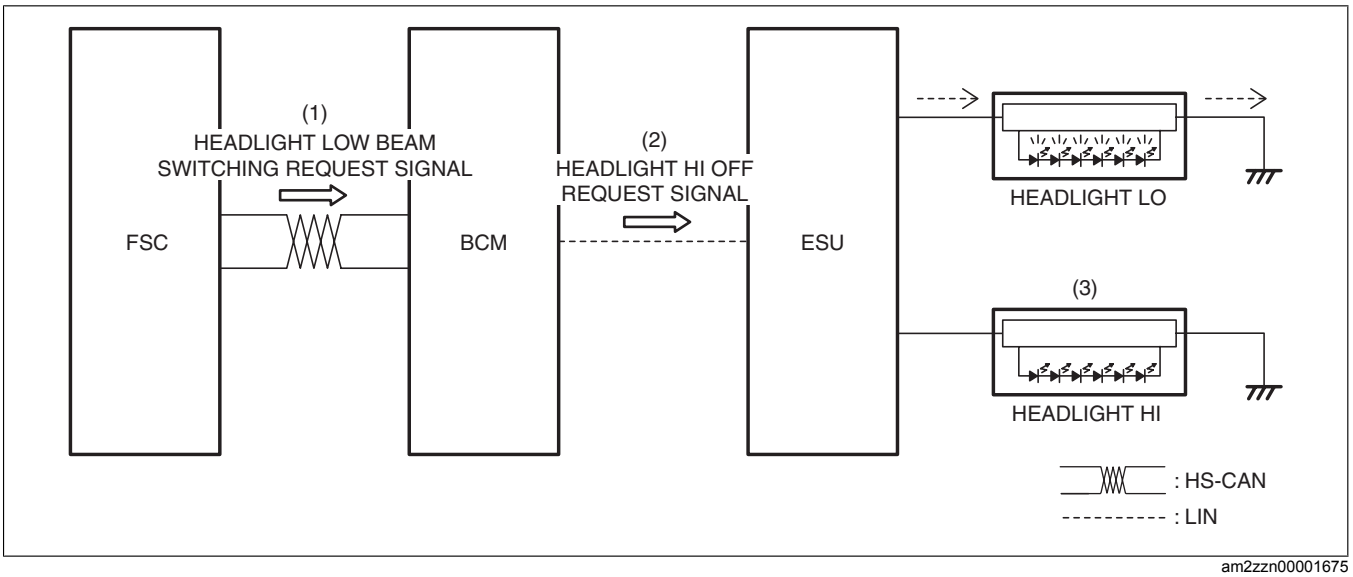
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Operation

Low beam switching operation

- With the light switch in the AUTO position and the dimmer switch in the HI position and when the headlights are turned on, the headlights are switched to low beams when any of the following conditions are met:
  Operation conditions
  — Vehicles ahead, street lights (driving in town/city) are recognized
  — Vehicle speed is **20 km/h {12 mph} or less** (vehicle speed signal of **20 km/h {12 mph} or less** is received)

1. When any of the operation conditions is met, the forward sensing camera (FSC) sends (1) a headlight low beam switching request signal to the BCM via CAN communication.
2. When the BCM receives the headlight low beam switching request signal, it sends (2) the headlight HI off request signal to the electrical supply unit (ESU) via LIN communication.
3. When the electrical supply unit (ESU) receives the headlight HI off request signal, it turns the headlights HI off (3).



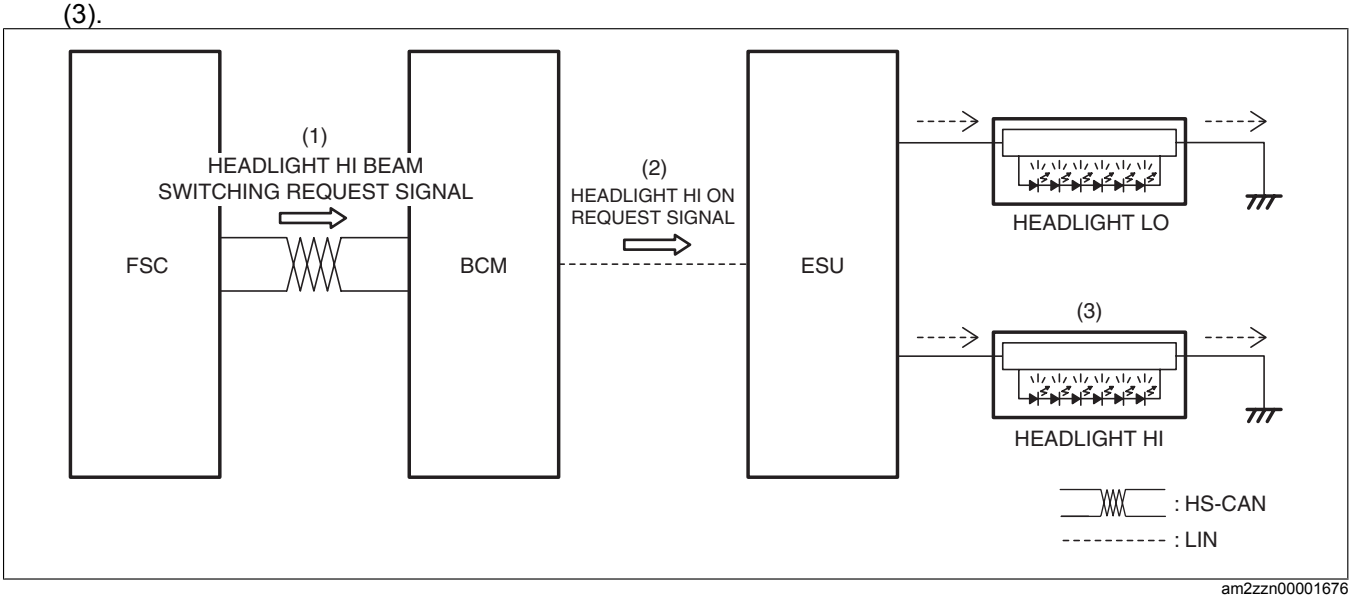
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High beam return operation

- When the following condition is met, the headlights are switched to high beams.
  Operation Conditions
  — Vehicle speed is **30 km/h {19 mph} or more** (vehicle signal of **30 km/h {19 mph} or more** is received) and vehicles ahead or street lights (driving in town/city) are not recognized

On-coming vehicles

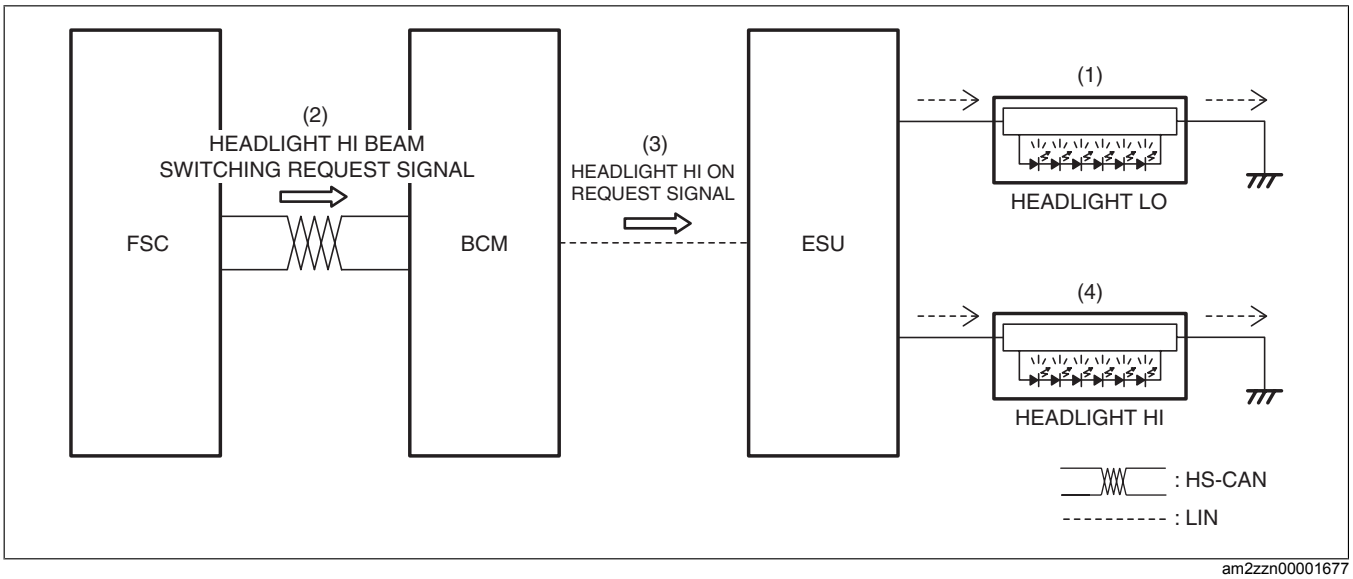
1. When the forward sensing camera (FSC) determines that a recognized on-coming vehicle moves away from the camera range, it sends (1) a headlight high beam switching request signal to the BCM via CAN communication.
2. When the BCM receives the headlight high beam switching request signal, it sends (2) the headlight HI on request signal to the electrical supply unit (ESU) via LIN communication.
3. When the electrical supply unit (ESU) receives the headlight HI on request signal, it turns the headlights HI on (3).



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Vehicles ahead or street lights (driving in town/city)

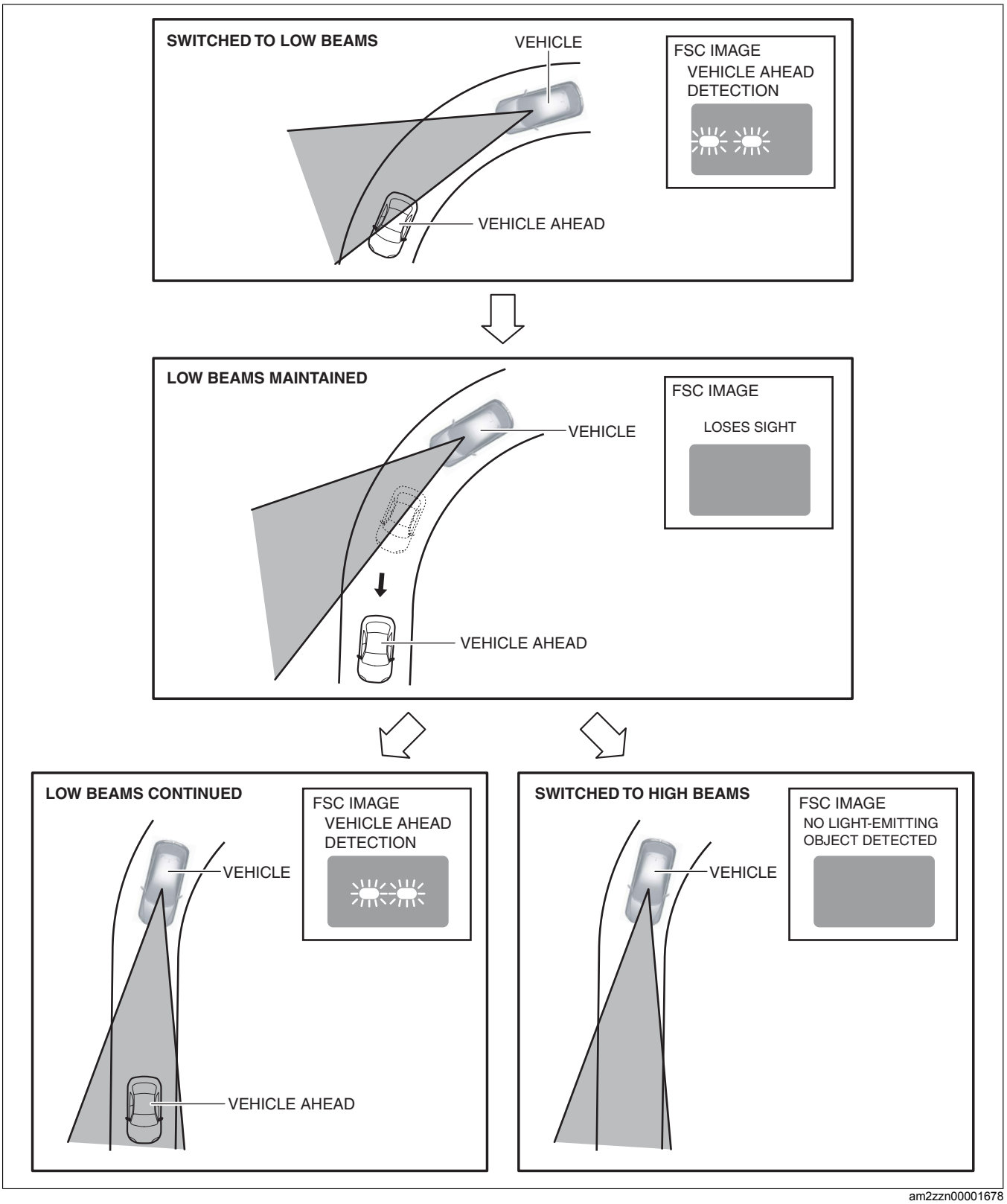
1. If the forward sensing camera (FSC) no longer detect a recognized vehicle ahead moves away from the camera range, or street lights (driving in town/city) in the camera range, it maintains (1) low beams for a certain period of time.
2. If the forward sensing camera (FSC) does not detect vehicles ahead or street lights (driving in town/city) after the certain period of time has elapsed, it sends (2) a headlight high beam switching request signal to the BCM via CAN communication.
3. When the BCM receives the headlight high beam switching request signal, it sends (3) the headlight HI on request signal to the electrical supply unit (ESU) via LIN communication.
4. When the electrical supply unit (ESU) receives the headlight HI on request signal, it turns the headlights HI on (4).



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Forward sensing camera (FSC) loses sight of vehicle ahead which is cornering

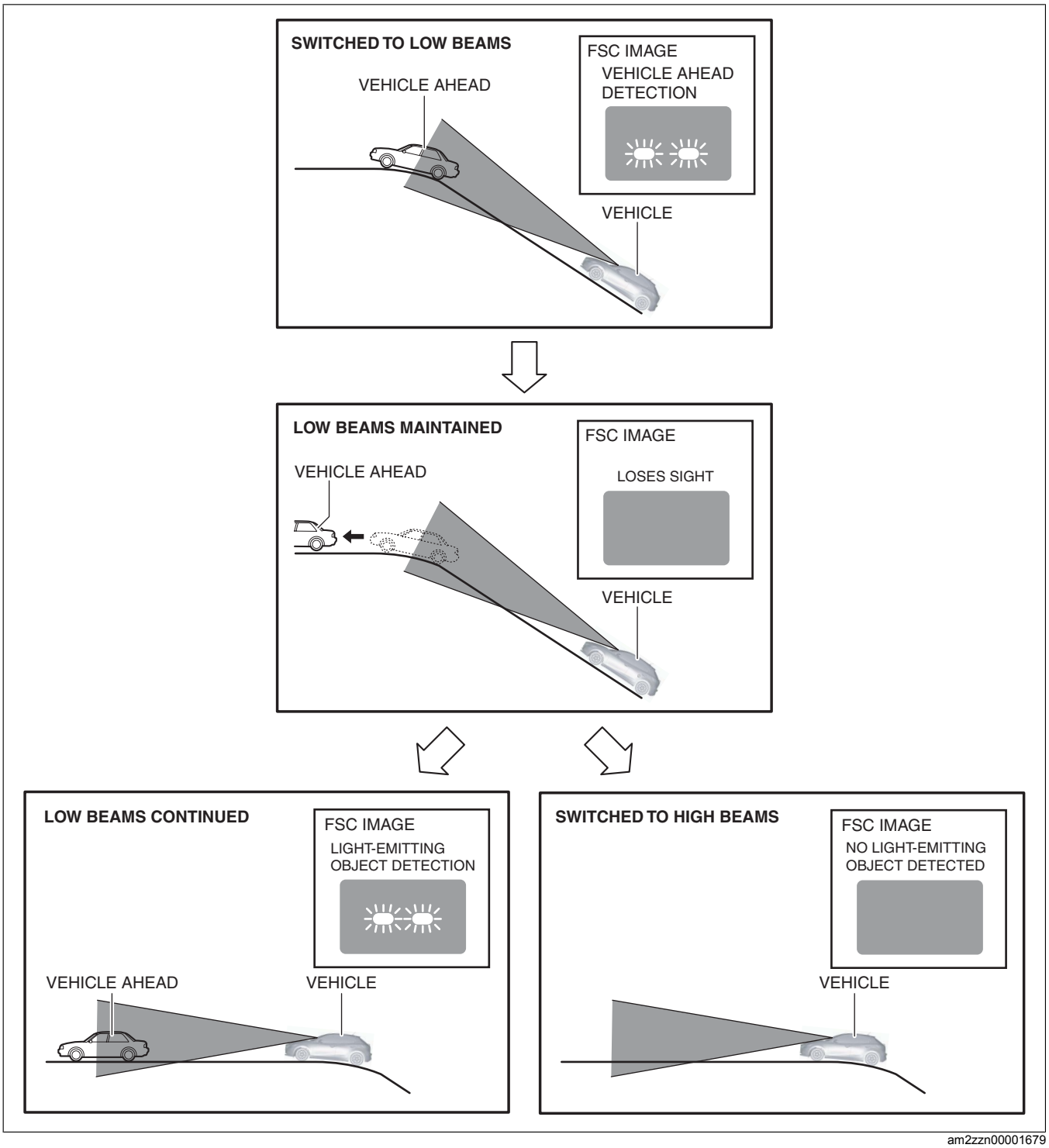
- If the forward sensing camera (FSC) loses sight of a recognized vehicle which is cornering, the forward sensing camera (FSC) maintains the low beams until reaching the locus at which the forward sensing camera (FSC) lost sight of it.
- If the vehicle ahead is not detected at the locus at which the forward sensing camera (FSC) had lost sight of it, the headlights are switched to high beams. If the vehicle ahead is detected, the low beams are maintained.



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Forward sensing camera (FSC) suddenly loses sight of light-emitting object (vehicles ahead, town/city streets (street lights))

- If the forward sensing camera (FSC) suddenly loses sight of a recognized light-emitting object in the camera range, the low beams are maintained for a certain period of time.
- Depending on whether a light-emitting object is detected or not after the certain period of time has elapsed, the headlights are switched to high beams or the system determines that the low beams are to be maintained.



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Fail-safe

- If a malfunction is detected in the system, control is inhibited and the headlights are switched to high beams.